report_32bit

April 08, 2025

Slides and markers

$AS/AS1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$AS/AS2_Les [1] CD11c CD123 CD279 CD3 CD4_2 CD4 CD8

$AS/AS3_Les [1] CD279 CD3 CD4 CD56 CD8

$CF/CF1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD8

$CF/CF2_Les [1] CD11c CD123 CD279 CD3 CD4 CD8

$CF/CF5_Les [1] CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$CJ/CJ1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$CJ/CJ2_Les [1] CD11c CD123 CD15 CD3 CD4 CD45 CD68 CD8

$DD/DD1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$DD/DD2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$DD/DD3_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$EB/EB_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD8

$JK/JK1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD56 CD68
[9] CD68_raw CD69 CD8

$JK/JK2_Les [1] CD3 CD4 CD8

$JK/JK3_Les [1] CD123 CD15 CD279 CD3 CD4 CD68 CD68_raw CD69
[9] CD8

$JM/JM1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$JM/JM2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$JM/JM3_Les [1] CD3 CD8

$JT/JT_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD8

$LD/LD_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$LG/LG1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD69 CD8

$LG/LG2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$LG/LG3_Les [1] CD3 CD4 CD68 CD68_raw CD8

$LM/LM_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$LS/LS1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$LS/LS2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56
[9] CD68 CD68_raw CD69 CD8

$LS/LS3_Les [1] CD3 CD4 CD68 CD68_raw CD8

$SK/SK1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD8

$SK/SK2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8

$SK/SK3_Les [1] CD11c CD123 CD279 CD3 CD4 CD68 CD68_raw CD8

Marker groups

- T cell: CD3 CD4 CD8 CD69 CD279 Foxp3
- Neutrophil: CD15
- Dendritic cells: CD11c CD123
- Macrophage: CD68
- NK cells: CD56. This marker is expressed on both NK and nerve fibers, need to check if the staining on the latter affects the NK cell count.

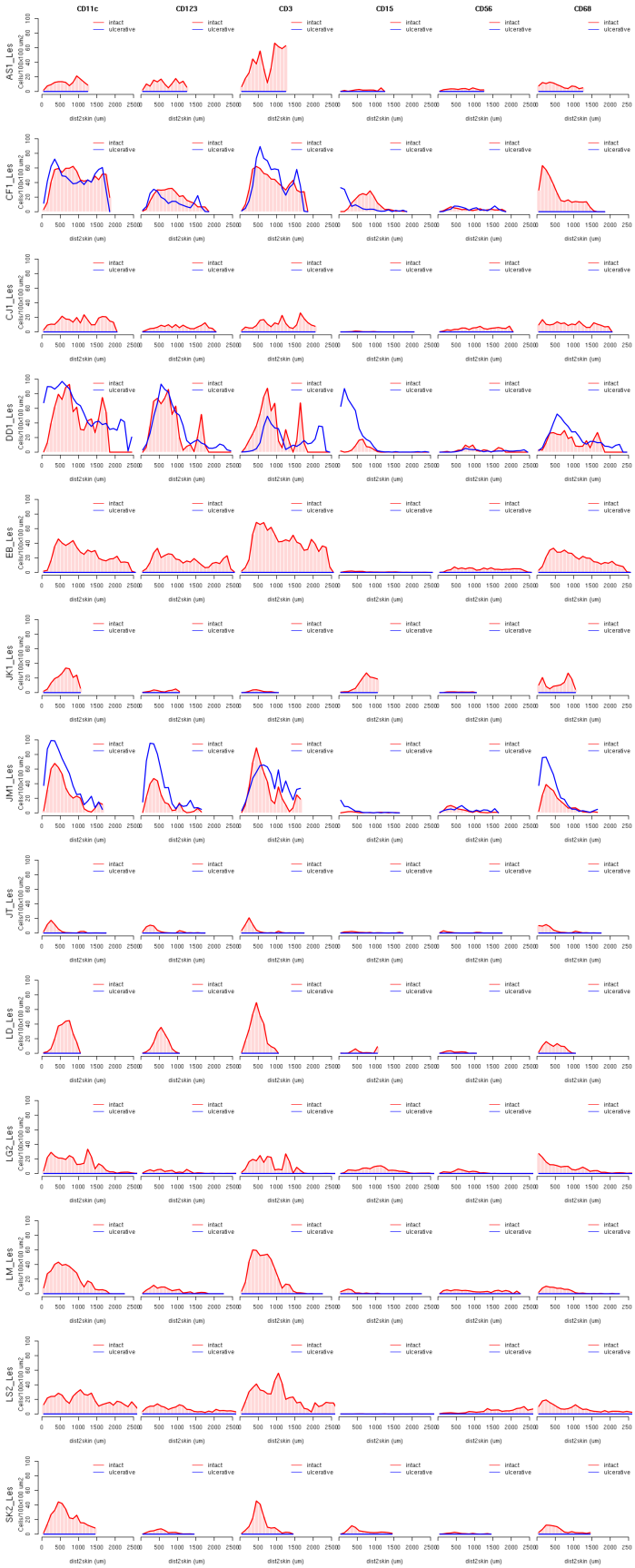
The key markers are: CD11c CD123 CD3 CD15 CD56 CD68. There are 13 lesion biopsies in total. We will focus on lesion biopsies with the key markers data.

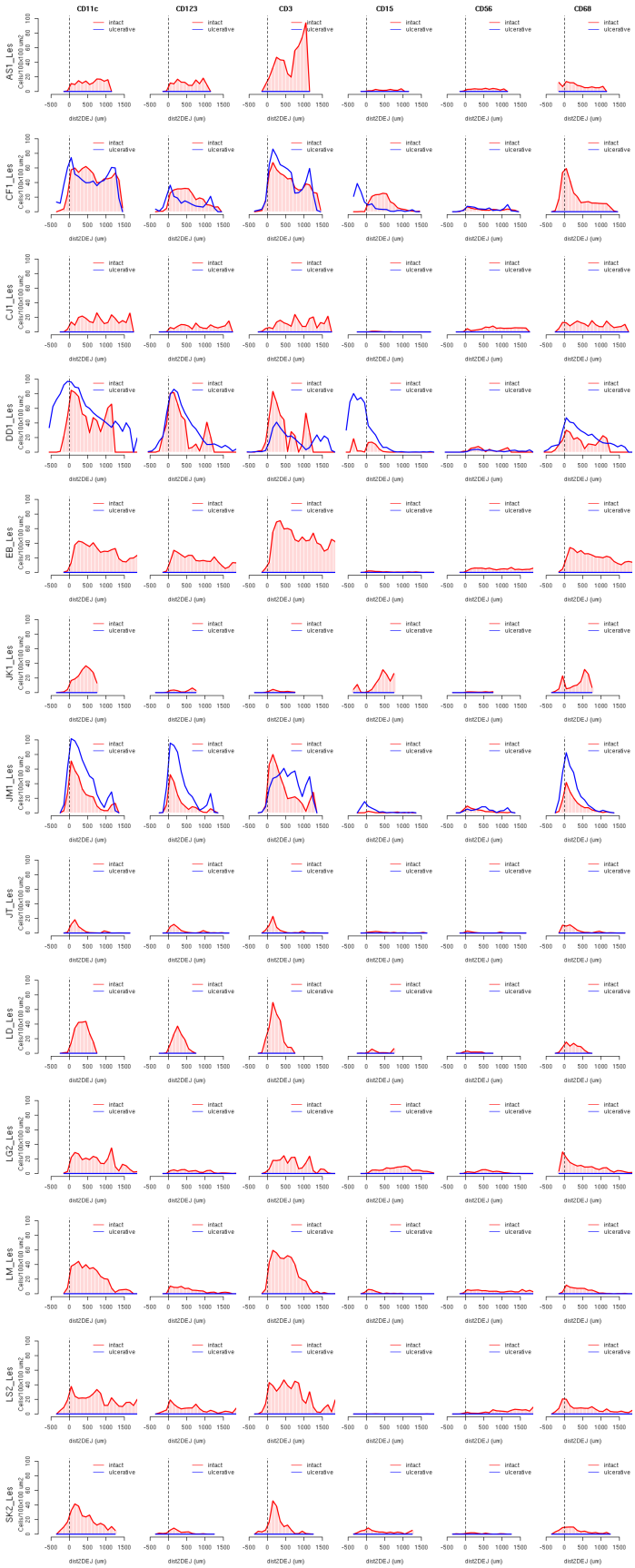
Some lesion biopsies have multiple sections. The following choices of slides are made: AS/AS1_Les, CF/CF1_Les, CJ/CJ1_Les, DD/DD1_Les, EB/EB_Les, JK/JK1_Les, JM/JM1_Les, JT/JT_Les,LD/LD_Les, LG/LG2_Les, LM/LM_Les, LS/LS2_Les, SK/SK2_Les. The only exceptions are

- For CF CD68, CF5 is used instead of CF1 for better quality.

Figures

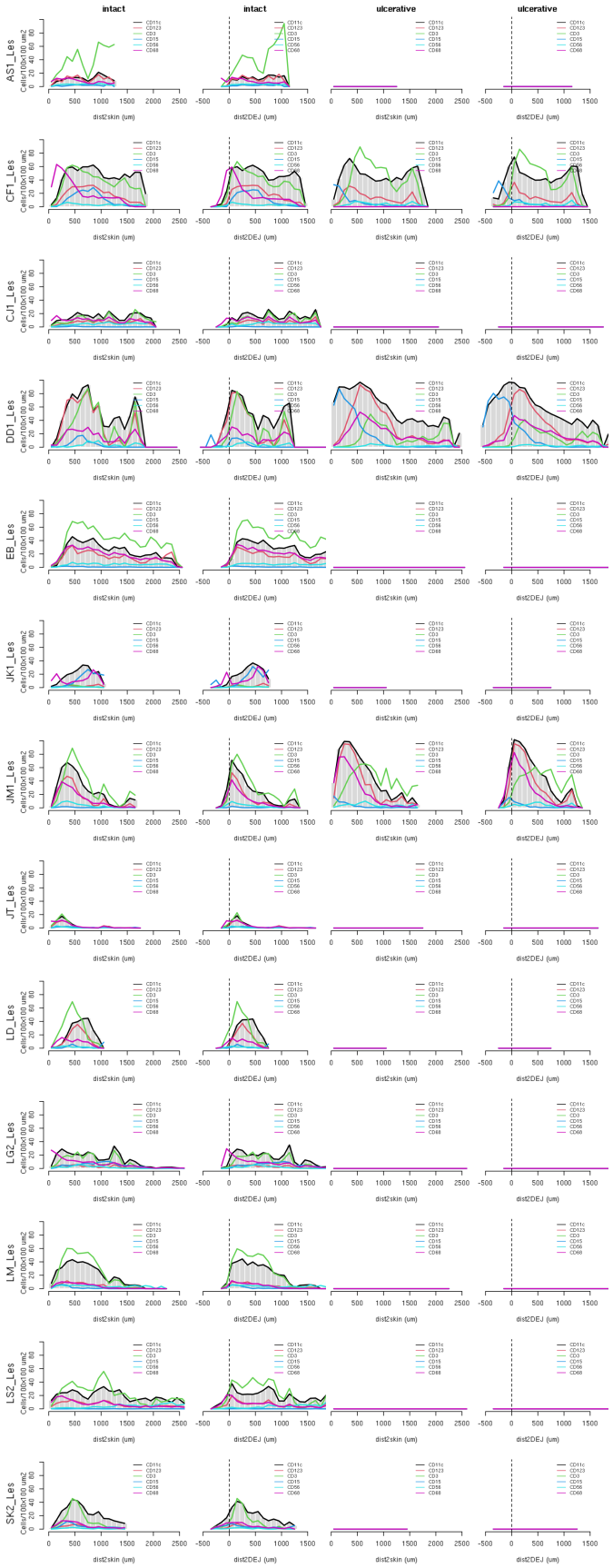
Number of cells per 100x100 um2 tissue as a function of distance to skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width.


CD11c
CD123
CD3
CD15
CD56
CD68
intact
ulcerative
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100x100 um2
dist2skin (um)


CD11c
CD123
CD3
CD15
CD56
CD68
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100x100 um2
dist2DEJ (um)
intact
ulcerative

Comparing the first two rows in the first column, we see that the ratio of the area under the curve across both epi and dermis between red and blue(ulcerative and intact) is about 1 for CD3/CF and 1/3 for CD3/DD. This suggests that CD3 is depleted in lesion in DD, but not in CF.

Compare lesion with normal in CD15/CF, we see that the lesion peak is shifted to the left compared to the normal peak. This pattern is more clear in CF than in DD. We don't see this pattern for CD3.


intact
intact
ulcerative
ulcerative
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
CD11c
CD123
CD3
CD15
CD56
CD68
dist2skin (um)
dist2DEJ (um)

That the CD15 dist2skin peak in DD/lesion is near skin while the corresponding dist2DEJ peak is near DEJ is due to the fact that the DEJ line is very curvy.

Correlation studies

Correlate shedding rate of 13 biopies with CD3. Explore different features:

- epi, dermis up to 500, demis up to 1000, total
- count, density, regularized density

Count tables

For each slide, there are two tables, one for up to 500 um below DEJ and one for up to 1000 um below DEJ.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	18	1372	0	0	90	1575			92	1329		
CD123	15	1353	0	0	75	1553			76	1266		
CD15	2	195	0	0	10	223			10	209		
CD279	69	1453	0	0	348	1668			354	1669		
CD3	187	4071	0	0	944	4673			959	4470		
CD4	92	3237	0	0	464	3716			472	3652		
CD45	337	6773	0	0	1701	7775			1016	6842		
CD56	12	363	0	0	60	416			61	333		
CD68	134	1037	0	0	676	1190			950	743		
CD68_raw	247	933	0	0	1246	1071			1088	634		
CD69	233	1917	0	0	1176	2200			1195	2060		
CD8	114	1343	0	0	575	1541			585	1471		
area	0.20	0.87	0.00	0.00								

Table 1: AS1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	18	881	0	0	90	1728			92	1249		
CD123	15	938	0	0	75	1840			76	1314		
CD15	2	116	0	0	10	227			10	182		
CD279	69	810	0	0	348	1589			354	1298		
CD3	187	2536	0	0	944	4976			959	4142		
CD4	92	1934	0	0	464	3795			472	3161		
CD45	337	4323	0	0	1701	8483			1016	6438		
CD56	12	235	0	0	60	461			61	329		
CD68	134	840	0	0	676	1648			950	994		
CD68_raw	247	766	0	0	1246	1503			1088	875		
CD69	233	1163	0	0	1176	2282			1195	1758		
CD8	114	808	0	0	575	1585			585	1258		
area	0.20	0.51	0.00	0.00								

Table 2: AS1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1396	7069	0	0	2052	3499			1230	2114		
CD123	605	3119	0	0	889	1543			511	868		
CD279	624	3994	0	0	917	1976			759	1308		
CD3	1284	9955	0	0	1887	4927			1175	3204		
CD4_2	1635	7638	0	0	2403	3780			2004	2490		
CD4	2081	8021	0	0	3059	3970			2279	2503		
CD8	819	3921	0	0	1204	1940			768	1214		
area	0.68	2.02	0.00	0.00								

Table 3: AS2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1396	5801	0	0	2052	4838			1230	2798		
CD123	605	2630	0	0	889	2193			511	1184		
CD279	624	3200	0	0	917	2669			759	1725		
CD3	1284	7995	0	0	1887	6668			1175	4255		
CD4_2	1635	6067	0	0	2403	5060			2004	3245		
CD4	2081	6450	0	0	3059	5380			2279	3272		
CD8	819	3167	0	0	1204	2641			768	1563		
area	0.68	1.20	0.00	0.00								

Table 4: AS2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD279	327	3620	0	0	569	1857			371	1318		
CD3	578	10861	0	0	1005	5574			630	4091		
CD4	302	9296	0	0	525	4771			331	3678		
CD56	77	1416	0	0	133	726			98	512		
CD8	538	4858	0	0	936	2493			583	1712		
area	0.57	1.95	0.00	0.00								

Table 5: AS3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD279	327	2753	0	0	569	2513			371	1795		
CD3	578	7960	0	0	1005	7266			630	5220		
CD4	302	6627	0	0	525	6049			331	4609		
CD56	77	925	0	0	133	844			98	447		
CD8	538	3577	0	0	936	3265			583	2081		
area	0.57	1.10	0.00	0.00								

Table 6: AS3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	631	7926	1505	4139	1443	6261	3886	5829	969	5224	3057	4268
CD123	124	3900	407	1457	283	3080	1051	2052	196	2593	787	1220
CD15	8	2460	934	403	18	1943	2412	567	21	1685	2509	324
CD279	159	1622	59	1249	363	1281	152	1759	248	1023	117	1314
CD3	405	7078	277	5041	926	5591	715	7100	595	4572	519	5365
CD4	334	5588	216	3488	764	4414	557	4912	498	3717	500	3914
CD45	540	8700	995	5538	1235	6872	2569	7800	799	5711	1849	5816
CD56	64	529	36	404	146	417	92	569	135	294	126	386
CD68	1576	3700	0	0	3605	2922	0	0	2433	1821		
CD8	296	3668	210	2990	677	2897	542	4211	698	2292	420	3144
area	0.44	1.27	0.39	0.71								

Table 7: CF1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	631	4834	1505	2826	1443	7900	3886	7368	969	5872	3057	4567
CD123	124	2464	407	1165	283	4026	1051	3037	196	3114	787	1658
CD15	8	1680	934	353	18	2745	2412	920	21	2331	2509	543
CD279	159	1134	59	905	363	1853	152	2359	248	1378	117	1719
CD3	405	4655	277	3728	926	7607	715	9720	595	5685	519	7216
CD4	334	3365	216	2385	764	5499	557	6218	498	4128	500	4642
CD45	540	5296	995	3915	1235	8655	2569	10207	799	6352	1849	6980
CD56	64	373	36	308	146	609	92	803	135	363	126	512
CD68	1576	2886	0	0	3605	4716	0	0	2433	2545		
CD8	296	2472	210	2217	677	4039	542	5780	698	2884	420	4215
area	0.44	0.61	0.39	0.38								

Table 8: CF1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1018	9592	752	1646	1861	6216	2566	5611	1279	4984	2021	4364
CD123	292	7591	234	783	534	4919	798	2669	506	3658	795	1827
CD279	189	2016	152	516	345	1306	518	1758	298	994	485	1113
CD3	330	5949	303	1632	603	3855	1034	5563	389	3081	968	3603
CD4	136	4208	148	1017	248	2727	505	3466	231	2380	473	3136
CD8	263	4535	261	1632	481	2939	890	5563	385	2388	814	4118
area	0.55	1.54	0.29	0.29								

Table 9: CF2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1018	5869	752	1282	1861	7357	2566	6062	1279	5025	2021	3683
CD123	292	5287	234	692	534	6627	798	3272	506	4684	795	2015
CD279	189	1504	152	459	345	1885	518	2170	298	1420	485	1417
CD3	330	4273	303	1465	603	5356	1034	6927	389	4244	968	4578
CD4	136	2685	148	810	248	3365	505	3830	231	2817	473	2980
CD8	263	3123	261	1383	481	3915	890	6539	385	3052	814	4546
area	0.55	0.80	0.29	0.21								

Table 10: CF2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	698	3776	0	0	1901	1970			1393	1179		
CD15	614	3611	0	0	1672	1884			947	1450		
CD279	85	2577	0	0	231	1344			174	1067		
CD3	263	5572	0	0	716	2907			527	2465		
CD4	505	10867	0	0	1375	5670			912	5032		
CD45	1514	12844	0	0	4123	6701			3343	5574		
CD56	1	279	0	0	2	145			3	140		
CD68	1576	3700	0	0	4292	1930			2784	1232		
CD69	256	2961	0	0	697	1545			457	1245		
CD8	200	1943	0	0	544	1013			345	881		
area	0.37	1.92	0.00	0.00								

Table 11: CF5_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	698	3522	0	0	1901	3809			1393	2338		
CD15	614	3102	0	0	1672	3355			947	2685		
CD279	85	1959	0	0	231	2118			174	1634		
CD3	263	3660	0	0	716	3958			527	3096		
CD4	505	6110	0	0	1375	6608			912	5315		
CD45	1514	7756	0	0	4123	8388			3343	6143		
CD56	1	146	0	0	2	157			3	144		
CD68	1576	2886	0	0	4292	3121			2784	1739		
CD69	256	1843	0	0	697	1993			457	1350		
CD8	200	1123	0	0	544	1214			345	886		
area	0.37	0.92	0.00	0.00								

Table 12: CF5_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	157	3201	0	0	306	1975			197	1689		
CD123	13	1440	0	0	25	888			32	759		
CD15	1	88	0	0	1	54			2	47		
CD279	141	841	0	0	275	518			201	433		
CD3	196	2168	0	0	382	1337			269	1243		
CD4	397	4852	0	0	775	2994			572	2441		
CD45	529	7523	0	0	1033	4642			733	3904		
CD56	14	866	0	0	27	534			34	475		
CD68	581	2248	0	0	1134	1387			894	1104		
CD68_raw	263	1226	0	0	513	756			417	617		
CD69	194	1736	0	0	378	1071			288	957		
CD8	44	779	0	0	85	480			57	459		
area	0.51	1.62	0.00	0.00								

Table 13: CJ1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	157	2005	0	0	306	2243			197	1741		
CD123	13	910	0	0	25	1018			32	802		
CD15	1	76	0	0	1	85			2	75		
CD279	141	564	0	0	275	631			201	496		
CD3	196	1223	0	0	382	1368			269	1165		
CD4	397	3153	0	0	775	3528			572	2576		
CD45	529	4751	0	0	1033	5316			733	3988		
CD56	14	405	0	0	27	453			34	285		
CD68	581	1492	0	0	1134	1669			894	1194		
CD68_raw	263	813	0	0	513	909			417	691		
CD69	194	1062	0	0	378	1188			288	990		
CD8	44	427	0	0	85	477			57	424		
area	0.51	0.89	0.00	0.00								

Table 14: CJ1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	323	4247	0	0	494	2246			331	1792		
CD123	56	1739	0	0	85	919			110	777		
CD15	3	28	0	0	4	14			5	14		
CD3	36	2392	0	0	55	1265			40	1170		
CD4	60	2626	0	0	91	1388			71	1274		
CD45	489	6667	0	0	749	3526			517	2906		
CD68	399	2450	0	0	611	1295			431	980		
CD8	46	805	0	0	70	425			57	404		
area	0.65	1.89	0.00	0.00								

Table 15: CJ2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	323	2937	0	0	494	2760			331	2078		
CD123	56	1171	0	0	85	1100			110	899		
CD15	3	23	0	0	4	21			5	18		
CD3	36	1508	0	0	55	1417			40	1319		
CD4	60	1688	0	0	91	1586			71	1478		
CD45	489	4462	0	0	749	4193			517	3244		
CD68	399	1820	0	0	611	1710			431	1272		
CD8	46	474	0	0	70	445			57	410		
area	0.65	1.06	0.00	0.00								

Table 16: CJ2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	360	2203	10865	17752	3096	6651	8365	7851	2401	4971	7292	6302
CD123	308	1599	3062	14146	2649	4827	2357	6256	2086	3143	1621	4903
CD15	12	213	9301	3522	103	643	7161	1557	545	505	6561	942
CD279	17	594	288	3275	146	1793	221	1448	318	1398	169	1329
CD3	85	1535	140	6126	731	4634	107	2709	580	3866	81	2435
CD4	233	2127	443	11429	2004	6421	341	5054	1608	4679	339	4463
CD45	451	2935	12474	20869	3879	8861	9604	9230	2844	6688	8460	7457
CD56	7	134	3	612	60	404	2	270	131	377	8	252
CD68	141	673	1818	8087	1212	2031	1399	3576	1013	1467	1025	2814
CD69	19	253	31	2283	163	763	23	1009	356	708	23	961
CD8	37	634	75	2692	318	1914	57	1190	248	1760	50	1096
area	0.12	0.33	1.30	2.26								

Table 17: DD1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	360	1551	10865	11760	3096	9814	8365	10881	2401	6493	7292	7866
CD123	308	1463	3062	10459	2649	9257	2357	9677	2086	6049	1621	7179
CD15	12	211	9301	3230	103	1335	7161	2988	545	743	6561	1822
CD279	17	512	288	2289	146	3239	221	2118	318	2590	169	1964
CD3	85	1308	140	4119	731	8276	107	3811	580	6177	81	3378
CD4	233	1648	443	7342	2004	10427	341	6793	1608	7138	339	5739
CD45	451	2153	12474	13998	3879	13623	9604	12952	2844	9419	8460	9545
CD56	7	101	3	381	60	639	2	352	131	613	8	325
CD68	141	497	1818	5491	1212	3144	1399	5080	1013	2014	1025	3602
CD69	19	216	31	1188	163	1366	23	1099	356	1170	23	1019
CD8	37	503	75	1685	318	3182	57	1559	248	2633	50	1410
area	0.12	0.16	1.30	1.08								

Table 18: DD1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	875	1740	1003	7548	3036	2783	1294	2853	2351	2450	788	2240
CD123	266	743	759	5397	923	1188	979	2040	759	775	486	988
CD15	12	395	2932	3889	41	631	3784	1470	46	428	5368	387
CD279	27	40	37	128	93	63	47	48	75	60	32	35
CD3	293	2213	449	6549	1016	3540	579	2475	887	2546	281	1463
CD4	241	1276	313	4330	836	2041	404	1636	685	1479	176	1060
CD45	1148	4987	3613	22440	3984	7979	4664	8482	3422	6085	3149	5336
CD56	30	88	72	449	104	140	92	169	88	112	112	125
CD68	162	249	413	1451	562	398	533	548	457	238	406	313
CD69	530	1530	578	5345	1839	2447	746	2020	1484	1946	898	1210
CD8	124	615	129	1683	430	983	166	636	350	664	104	391
area	0.29	0.63	0.77	2.65								

Table 19: DD2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	875	831	1003	5243	3036	3224	1294	3477	2351	2634	788	2496
CD123	266	554	759	5097	923	2149	979	3381	759	1139	486	1922
CD15	12	381	2932	3850	41	1478	3784	2553	46	916	5368	832
CD279	27	33	37	111	93	128	47	73	75	82	32	53
CD3	293	1634	449	5844	1016	6341	579	3876	887	3769	281	2520
CD4	241	891	313	3654	836	3457	404	2423	685	2055	176	1663
CD45	1148	3160	3613	17682	3984	12262	4664	11729	3422	7571	3149	6804
CD56	30	44	72	343	104	170	92	227	88	106	112	176
CD68	162	184	413	1207	562	714	533	800	457	329	406	447
CD69	530	1011	578	4772	1839	3923	746	3165	1484	2596	898	2130
CD8	124	469	129	1473	430	1820	166	977	350	980	104	657
area	0.29	0.26	0.77	1.51								

Table 20: DD2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	478	2859	10260	17020	2745	6305	8695	6913	1731	4710	7851	4938
CD123	458	2163	3003	13393	2630	4770	2545	5440	1587	3028	1527	3271
CD15	46	529	8776	3516	264	1166	7437	1428	357	909	6834	661
CD279	22	258	186	1580	126	568	157	641	118	522	149	479
CD3	97	1086	355	3700	557	2395	300	1502	540	1939	199	1191
CD4	372	2498	598	12097	2136	5509	506	4913	1959	4081	262	3912
CD45	509	3197	11142	17714	2923	7050	9442	7195	1863	5144	8738	5118
CD56	18	159	41	669	103	350	34	271	104	327	84	233
CD68	344	1487	2300	10029	1975	3279	1949	4073	1678	2539	1174	2927
CD69	2	63	96	388	11	138	81	157	14	201	198	135
CD8	61	722	211	2741	350	1592	178	1113	345	1403	102	938
area	0.17	0.45	1.18	2.46								

Table 21: DD3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	478	1912	10260	12619	2745	10028	8695	9777	1731	6110	7851	6435
CD123	458	1902	3003	12031	2630	9975	2545	9321	1587	5585	1527	5968
CD15	46	524	8776	3470	264	2748	7437	2688	357	1568	6834	1441
CD279	22	228	186	1228	126	1195	157	951	118	1033	149	710
CD3	97	935	355	2563	557	4903	300	1985	540	3656	199	1463
CD4	372	1870	598	8248	2136	9807	506	6390	1959	6233	262	4710
CD45	509	2356	11142	13171	2923	12356	9442	10205	1863	7597	8738	6699
CD56	18	101	41	413	103	529	34	320	104	461	84	252
CD68	344	922	2300	7518	1975	4835	1949	5825	1678	3032	1174	3933
CD69	2	53	96	288	11	277	81	223	14	342	198	205
CD8	61	551	211	1677	350	2889	178	1299	345	2356	102	966
area	0.17	0.19	1.18	1.29								

Table 22: DD3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	56	5795	0	0	177	3802			128	3630		
CD123	47	3656	0	0	148	2398			99	2195		
CD15	9	178	0	0	28	116			36	88		
CD279	45	203	0	0	142	133			181	119		
CD3	143	8872	0	0	452	5820			575	5653		
CD4	221	10807	0	0	699	7090			525	6614		
CD45	391	14245	0	0	1236	9346			807	8633		
CD56	5	825	0	0	15	541			15	510		
CD68	121	4464	0	0	382	2928			275	2669		
CD8	62	3844	0	0	196	2522			184	2457		
area	0.32	1.52	0.00	0.00								

Table 23: EB_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	56	2940	0	0	177	4463			128	4014		
CD123	47	2025	0	0	148	3074			99	2564		
CD15	9	129	0	0	28	195			36	126		
CD279	45	121	0	0	142	183			181	149		
CD3	143	4500	0	0	452	6832			575	6372		
CD4	221	5758	0	0	699	8742			525	7543		
CD45	391	7493	0	0	1236	11376			807	9640		
CD56	5	426	0	0	15	646			15	568		
CD68	121	2445	0	0	382	3712			275	3075		
CD8	62	1852	0	0	196	2811			184	2653		
area	0.32	0.66	0.00	0.00								

Table 24: EB_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	255	3684	0	0	309	3277			195	2621		
CD123	22	435	0	0	26	386			22	293		
CD15	74	1738	0	0	89	1546			411	1913		
CD279	22	97	0	0	26	86			31	51		
CD3	26	391	0	0	31	347			50	188		
CD4	69	1659	0	0	83	1475			63	1292		
CD56	6	148	0	0	7	131			11	91		
CD68	1321	1568	0	0	1601	1394			994	1512		
CD68_raw	1701	2379	0	0	2062	2116			1300	2067		
CD69	33	165	0	0	40	146			64	97		
CD8	24	110	0	0	29	97			46	61		
area	0.82	1.12	0.00	0.00								

Table 25: JK1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	255	3235	0	0	309	3340			195	2742		
CD123	22	367	0	0	26	378			22	199		
CD15	74	1401	0	0	89	1446			411	1678		
CD279	22	93	0	0	26	96			31	53		
CD3	26	370	0	0	31	382			50	246		
CD4	69	1403	0	0	83	1448			63	1264		
CD56	6	132	0	0	7	136			11	82		
CD68	1321	1151	0	0	1601	1188			994	1035		
CD68_raw	1701	1851	0	0	2062	1911			1300	1536		
CD69	33	156	0	0	40	161			64	93		
CD8	24	110	0	0	29	113			46	61		
area	0.82	0.97	0.00	0.00								

Table 26: JK1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	18	417	0	0	23	373			32	256		
CD4	86	1748	0	0	112	1563			108	952		
CD8	13	97	0	0	17	86			13	53		
area	0.76	1.12	0.00	0.00								

Table 27: JK2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	18	334	0	0	23	390			32	282		
CD4	86	1484	0	0	112	1735			108	1106		
CD8	13	95	0	0	17	111			13	57		
area	0.76	0.86	0.00	0.00								

Table 28: JK2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	17	459	0	0	23	462			18	428		
CD15	7	717	0	0	9	722			9	876		
CD279	17	146	0	0	23	147			35	95		
CD3	20	251	0	0	27	253			42	161		
CD4	68	1422	0	0	94	1433			77	737		
CD68	756	1795	0	0	1045	1809			627	1053		
CD68_raw	1263	1499	0	0	1745	1511			1005	949		
CD69	22	158	0	0	30	159			30	133		
CD8	15	56	0	0	20	56			31	28		
area	0.72	0.99	0.00	0.00								

Table 29: JK3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	17	372	0	0	23	461			18	256		
CD15	7	499	0	0	9	618			9	606		
CD279	17	127	0	0	23	157			35	82		
CD3	20	224	0	0	27	277			42	195		
CD4	68	1316	0	0	94	1631			77	1022		
CD68	756	1595	0	0	1045	1977			627	1214		
CD68_raw	1263	1311	0	0	1745	1625			1005	1005		
CD69	22	133	0	0	30	164			30	72		
CD8	15	52	0	0	20	64			31	26		
area	0.72	0.81	0.00	0.00								

Table 30: JK3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	929	5747	1199	7120	1912	5058	4988	8169	1496	2536	3646	5181
CD123	376	3507	649	5802	773	3086	2700	6656	580	1276	1783	3789
CD15	6	122	326	345	12	107	1356	395	11	49	1156	179
CD279	182	2440	62	1946	374	2147	257	2232	284	951	337	1583
CD3	336	6967	155	4928	691	6131	644	5654	533	3599	458	4804
CD4	621	7178	448	5687	1278	6317	1863	6525	991	3581	1365	5082
CD45	1238	11869	1638	10638	2548	10446	6814	12205	2000	6818	5564	9402
CD56	152	670	44	495	312	589	183	567	232	270	239	442
CD68	623	2813	1122	4096	1282	2475	4667	4699	1004	1057	3658	2386
CD68_raw	860	2235	931	3089	1770	1967	3873	3544	1399	822	3200	1808
CD69	731	7062	898	7673	1504	6215	3736	8803	1135	2986	2534	5725
CD8	176	2236	62	1565	362	1967	257	1795	275	1012	174	1489
area	0.49	1.14	0.24	0.87								

Table 31: JM1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	929	5124	1199	5925	1912	7690	4988	11805	1496	4155	3646	8043
CD123	376	3271	649	5209	773	4909	2700	10378	580	2259	1783	6730
CD15	6	103	326	333	12	154	1356	663	11	46	1156	319
CD279	182	2342	62	1678	374	3514	257	3343	284	1886	337	2771
CD3	336	6090	155	3206	691	9139	644	6387	533	5788	458	5279
CD4	621	6232	448	4008	1278	9352	1863	7985	991	5593	1365	6089
CD45	1238	9412	1638	7482	2548	14125	6814	14907	2000	8955	5564	11006
CD56	152	615	44	344	312	922	183	685	232	467	239	532
CD68	623	2592	1122	3759	1282	3890	4667	7489	1004	1818	3658	4369
CD68_raw	860	2071	931	2864	1770	3108	3873	5706	1399	1435	3200	3403
CD69	731	6708	898	6397	1504	10067	3736	12745	1135	5825	2534	9086
CD8	176	2039	62	994	362	3060	257	1980	275	1757	174	1580
area	0.49	0.67	0.24	0.50								

Table 32: JM1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1561	4294	0	0	2517	4465			1551	1633		
CD123	298	2103	0	0	480	2187			339	816		
CD15	12	76	0	0	19	79			76	31		
CD279	161	511	0	0	259	531			181	264		
CD3	403	4870	0	0	649	5064			456	1525		
CD4	656	5371	0	0	1058	5585			755	2262		
CD45	798	4536	0	0	1287	4717			959	1369		
CD56	188	678	0	0	303	705			212	234		
CD68	848	1857	0	0	1367	1931			897	770		
CD68_raw	2839	2193	0	0	4578	2280			2993	699		
CD69	791	6050	0	0	1275	6291			883	1983		
CD8	211	2087	0	0	340	2170			230	645		
area	0.62	0.96	0.00	0.00								

Table 33: JM2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1561	4161	0	0	2517	5588			1551	2895		
CD123	298	2055	0	0	480	2760			339	1520		
CD15	12	71	0	0	19	95			76	25		
CD279	161	509	0	0	259	683			181	316		
CD3	403	4813	0	0	649	6464			456	3121		
CD4	656	5185	0	0	1058	6964			755	3849		
CD45	798	4499	0	0	1287	6042			959	2796		
CD56	188	666	0	0	303	894			212	405		
CD68	848	1817	0	0	1367	2440			897	1187		
CD68_raw	2839	2159	0	0	4578	2899			2993	1408		
CD69	791	5997	0	0	1275	8054			883	4114		
CD8	211	2063	0	0	340	2770			230	1243		
area	0.62	0.74	0.00	0.00								

Table 34: JM2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	233	4303	0	0	449	5757			346	2285		
CD8	170	1750	0	0	328	2341			255	859		
area	0.52	0.75	0.00	0.00								

Table 35: JM3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	233	4264	0	0	449	6414			346	3149		
CD8	170	1734	0	0	328	2608			255	1168		
area	0.52	0.66	0.00	0.00								

Table 36: JM3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	24	1182	0	0	107	781			107	480		
CD123	9	896	0	0	40	592			40	341		
CD15	4	218	0	0	17	144			17	104		
CD279	15	149	0	0	66	98			66	65		
CD3	92	1195	0	0	410	790			410	451		
CD4	72	1189	0	0	321	786			321	460		
CD45	161	3393	0	0	718	2243			718	1380		
CD56	8	174	0	0	35	115			35	58		
CD68	243	923	0	0	1084	610			1080	347		
CD68_raw	558	956	0	0	2489	632			2490	366		
CD8	54	518	0	0	240	342			241	222		
area	0.22	1.51	0.00	0.00								

Table 37: JT_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	24	1125	0	0	107	1311			107	856		
CD123	9	824	0	0	40	960			40	630		
CD15	4	179	0	0	17	208			17	165		
CD279	15	134	0	0	66	156			66	100		
CD3	92	1139	0	0	410	1328			410	907		
CD4	72	1105	0	0	321	1288			321	865		
CD45	161	3082	0	0	718	3594			718	2487		
CD56	8	155	0	0	35	180			35	79		
CD68	243	847	0	0	1084	987			1080	634		
CD68_raw	558	868	0	0	2489	1012			2490	642		
CD8	54	462	0	0	240	538			241	387		
area	0.22	0.86	0.00	0.00								

Table 38: JT_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	41	2501	0	0	118	4100			106	3434		
CD123	20	1599	0	0	58	2621			74	1948		
CD15	5	176	0	0	14	288			24	251		
CD279	89	440	0	0	258	721			190	480		
CD3	540	3111	0	0	1566	5100			1047	3340		
CD4	209	2014	0	0	606	3302			414	2345		
CD45	794	5665	0	0	2302	9288			1590	5659		
CD56	48	144	0	0	139	236			179	137		
CD68	233	862	0	0	675	1413			484	788		
CD69	392	2232	0	0	1136	3659			756	2287		
CD8	223	993	0	0	646	1628			832	993		
area	0.34	0.61	0.00	0.00								

Table 39: LD_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	41	2167	0	0	118	4715			106	4077		
CD123	20	1528	0	0	58	3324			74	2704		
CD15	5	165	0	0	14	359			24	260		
CD279	89	433	0	0	258	942			190	581		
CD3	540	2992	0	0	1566	6510			1047	4603		
CD4	209	1868	0	0	606	4064			414	2983		
CD45	794	5078	0	0	2302	11048			1590	7926		
CD56	48	143	0	0	139	311			179	167		
CD68	233	820	0	0	675	1784			484	1060		
CD69	392	2156	0	0	1136	4691			756	3631		
CD8	223	948	0	0	646	2062			832	1323		
area	0.34	0.46	0.00	0.00								

Table 40: LD_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1074	5057	942	5215	2325	3056	4586	5237	1760	2144	3771	4147
CD123	132	916	613	1498	285	553	2984	1504	202	272	2218	635
CD15	6	101	14	49	12	61	68	49	11	40	71	37
CD279	67	339	13	257	145	204	63	258	103	90	84	238
CD3	205	2000	83	1327	443	1208	404	1332	305	684	277	1150
CD4	303	2587	95	2189	656	1563	462	2198	452	964	336	1946
CD45	967	6473	798	5806	2093	3912	3885	5830	1525	2681	3093	4868
CD56	34	247	6	242	73	149	29	243	55	87	39	224
CD69	169	1166	73	1225	365	704	355	1230	280	367	248	1035
CD8	178	1663	69	1223	385	1005	335	1228	276	646	231	1053
area	0.46	1.65	0.21	1.00								

Table 41: LG1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1074	3962	942	3627	2325	4918	4586	8633	1760	3219	3771	5820
CD123	132	828	613	1376	285	1027	2984	3275	202	485	2218	1162
CD15	6	87	14	32	12	107	68	76	11	58	71	37
CD279	67	319	13	195	145	395	63	464	103	173	84	397
CD3	205	1731	83	946	443	2148	404	2251	305	1163	277	1744
CD4	303	2272	95	1509	656	2820	462	3591	452	1698	336	2868
CD45	967	5210	798	3952	2093	6467	3885	9407	1525	4174	3093	6849
CD56	34	224	6	161	73	278	29	383	55	140	39	325
CD69	169	1109	73	1001	365	1376	355	2382	280	740	248	1832
CD8	178	1348	69	892	385	1673	335	2123	276	1009	231	1642
area	0.46	0.81	0.21	0.42								

Table 42: LG1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	93	5966	0	0	274	2430			174	2084		
CD123	5	1047	0	0	14	426			15	364		
CD15	1	1619	0	0	2	659			3	656		
CD279	43	537	0	0	126	218			132	191		
CD3	58	4379	0	0	170	1783			178	1680		
CD4	92	5078	0	0	271	2068			174	1883		
CD45	194	9993	0	0	571	4070			329	3597		
CD56	8	874	0	0	23	355			24	321		
CD68	962	3235	0	0	2836	1317			1640	984		
CD68_raw	967	1731	0	0	2850	705			1520	510		
CD69	98	2628	0	0	288	1070			301	946		
CD8	40	2504	0	0	117	1019			123	947		
area	0.34	2.46	0.00	0.00								

Table 43: LG2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	93	3583	0	0	274	3081			174	2394		
CD123	5	691	0	0	14	594			15	477		
CD15	1	578	0	0	2	497			3	466		
CD279	43	349	0	0	126	300			132	250		
CD3	58	2497	0	0	170	2147			178	1968		
CD4	92	3054	0	0	271	2626			174	2286		
CD45	194	5971	0	0	571	5135			329	4223		
CD56	8	420	0	0	23	361			24	290		
CD68	962	2212	0	0	2836	1902			1640	1230		
CD68_raw	967	1200	0	0	2850	1032			1520	638		
CD69	98	1817	0	0	288	1562			301	1350		
CD8	40	1456	0	0	117	1252			123	1124		
area	0.34	1.16	0.00	0.00								

Table 44: LG2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	287	2999	149	2526	632	1716	472	2489	443	1070	344	1945
CD4	403	3396	241	3053	887	1944	763	3008	613	1341	482	2335
CD68	828	1401	1139	1512	1824	801	3608	1489	1471	516	2599	685
CD68_raw	1527	2002	1657	2178	3364	1146	5250	2146	2744	746	3852	1109
CD8	187	1791	97	1647	411	1025	307	1622	292	643	245	1337
area	0.45	1.75	0.32	1.01								

Table 45: LG3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	287	2600	149	1901	632	3163	472	4112	443	1868	344	2963
CD4	403	2796	241	2200	887	3401	763	4758	613	2214	482	3325
CD68	828	1113	1139	1276	1824	1354	3608	2760	1471	777	2599	1008
CD68_raw	1527	1565	1657	1752	3364	1904	5250	3789	2744	1097	3852	1532
CD8	187	1558	97	1199	411	1895	307	2593	292	1128	245	1995
area	0.45	0.82	0.32	0.46								

Table 46: LG3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	298	7681	0	0	745	4314			435	3401		
CD123	18	1632	0	0	45	916			49	656		
CD15	65	517	0	0	162	290			179	143		
CD279	39	177	0	0	97	99			107	68		
CD3	225	9871	0	0	563	5544			323	4415		
CD4	354	16036	0	0	885	9006			501	7293		
CD45	482	18467	0	0	1206	10372			713	8623		
CD56	45	926	0	0	112	520			124	393		
CD68	88	1563	0	0	220	877			243	585		
CD69	77	2900	0	0	192	1628			213	1373		
CD8	125	4096	0	0	312	2300			184	1921		
area	0.40	1.78	0.00	0.00								

Table 47: LM_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	298	5322	0	0	745	5360			435	4001		
CD123	18	1229	0	0	45	1237			49	860		
CD15	65	475	0	0	162	478			179	258		
CD279	39	134	0	0	97	134			107	74		
CD3	225	6814	0	0	563	6862			323	5313		
CD4	354	10892	0	0	885	10970			501	8489		
CD45	482	12185	0	0	1206	12272			713	9647		
CD56	45	658	0	0	112	662			124	467		
CD68	88	1212	0	0	220	1220			243	794		
CD69	77	1976	0	0	192	1990			213	1707		
CD8	125	2635	0	0	312	2653			184	2111		
area	0.40	0.99	0.00	0.00								

Table 48: LM_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1402	7533	0	0	2030	3501			1316	2359		
CD123	119	1401	0	0	172	651			111	445		
CD15	16	49	0	0	23	22			43	19		
CD279	496	4097	0	0	718	1904			440	1614		
CD3	692	10338	0	0	1002	4804			816	3979		
CD4	813	10287	0	0	1177	4781			1008	3674		
CD45	1692	14059	0	0	2450	6534			1442	5032		
CD56	133	1209	0	0	192	561			156	465		
CD68	908	1720	0	0	1315	799			789	453		
CD68_raw	1423	2347	0	0	2060	1090			1235	613		
CD69	1009	7850	0	0	1461	3648			1223	2780		
CD8	291	3429	0	0	421	1593			346	1304		
area	0.69	2.15	0.00	0.00								

Table 49: LS1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1402	5431	0	0	2030	4548			1316	2620		
CD123	119	978	0	0	172	819			111	466		
CD15	16	44	0	0	23	36			43	26		
CD279	496	2310	0	0	718	1934			440	1332		
CD3	692	6396	0	0	1002	5357			816	3873		
CD4	813	6816	0	0	1177	5709			1008	3818		
CD45	1692	9219	0	0	2450	7721			1442	5102		
CD56	133	636	0	0	192	532			156	295		
CD68	908	1309	0	0	1315	1096			789	489		
CD68_raw	1423	1807	0	0	2060	1513			1235	682		
CD69	1009	5192	0	0	1461	4348			1223	2873		
CD8	291	2117	0	0	421	1773			346	1248		
area	0.69	1.19	0.00	0.00								

Table 50: LS1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1567	7637	0	0	1490	3340			1098	2379		
CD123	477	3198	0	0	453	1399			343	866		
CD15	6	30	0	0	5	13			6	12		
CD279	342	1547	0	0	325	676			201	487		
CD3	980	11077	0	0	932	4845			562	3718		
CD4	1229	12721	0	0	1168	5565			714	4159		
CD45	1941	15205	0	0	1845	6651			1247	5057		
CD56	72	644	0	0	68	281			59	254		
CD68	1610	3072	0	0	1531	1343			1051	791		
CD68_raw	3474	4839	0	0	3303	2116			2445	1260		
CD69	652	5312	0	0	620	2323			373	1757		
CD8	302	2926	0	0	287	1280			252	967		
area	1.05	2.29	0.00	0.00								

Table 51: LS2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1567	5051	0	0	1490	4010			1098	2235		
CD123	477	2355	0	0	453	1870			343	937		
CD15	6	27	0	0	5	21			6	20		
CD279	342	1072	0	0	325	851			201	518		
CD3	980	7326	0	0	932	5817			562	3845		
CD4	1229	8433	0	0	1168	6696			714	4209		
CD45	1941	9942	0	0	1845	7894			1247	5046		
CD56	72	275	0	0	68	218			59	111		
CD68	1610	2329	0	0	1531	1849			1051	889		
CD68_raw	3474	3626	0	0	3303	2879			2445	1380		
CD69	652	3576	0	0	620	2839			373	1862		
CD8	302	1897	0	0	287	1506			252	931		
area	1.05	1.26	0.00	0.00								

Table 52: LS2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	643	11896	0	0	646	5140			384	4015		
CD4	797	13016	0	0	800	5624			502	4207		
CD68	2794	3479	0	0	2807	1503			1867	830		
CD68_raw	4129	4884	0	0	4149	2110			2945	1204		
CD8	319	3956	0	0	320	1709			180	1314		
area	1.00	2.31	0.00	0.00								

Table 53: LS3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	643	7999	0	0	646	6209			384	4349		
CD4	797	8700	0	0	800	6753			502	4246		
CD68	2794	2721	0	0	2807	2112			1867	942		
CD68_raw	4129	3720	0	0	4149	2887			2945	1291		
CD8	319	2507	0	0	320	1946			180	1207		
area	1.00	1.29	0.00	0.00								

Table 54: LS3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	84	2383	3133	4770	909	3323	3948	4900	564	2169	3401	1975
CD123	37	1361	1582	2848	400	1898	1993	2925	425	1177	1202	982
CD15	18	475	1775	1494	194	662	2237	1534	206	426	3762	627
CD279	43	791	545	1384	465	1103	686	1421	493	930	417	395
CD3	69	1063	717	1686	746	1482	903	1732	792	1200	472	625
CD4	30	952	421	1960	324	1327	530	2013	344	1003	267	625
CD45	199	3631	4090	7433	2154	5064	5155	7636	2285	3234	4453	3168
CD56	6	262	41	380	64	365	51	390	68	385	45	167
CD8	36	461	315	965	389	642	397	991	413	457	272	277
area	0.09	0.72	0.79	0.97								

Table 55: SK1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	84	2077	3133	4329	909	5258	3948	6156	564	3919	3401	2159
CD123	37	1278	1582	2642	400	3235	1993	3757	425	2381	1202	1042
CD15	18	441	1775	1418	194	1116	2237	2016	206	751	3762	905
CD279	43	773	545	1311	465	1957	686	1864	493	1575	417	440
CD3	69	1046	717	1548	746	2648	903	2201	792	2049	472	513
CD4	30	928	421	1850	324	2349	530	2631	344	1934	267	700
CD45	199	3257	4090	6726	2154	8245	5155	9565	2285	6036	4453	3418
CD56	6	254	41	350	64	643	51	497	68	554	45	175
CD8	36	450	315	919	389	1139	397	1306	413	878	272	275
area	0.09	0.39	0.79	0.70								

Table 56: SK1_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1024	7035	0	0	1491	3216			1071	2346		
CD123	96	945	0	0	139	432			129	271		
CD15	344	1075	0	0	500	491			395	289		
CD279	136	827	0	0	198	378			135	243		
CD3	343	4600	0	0	499	2103			422	1512		
CD4	580	5479	0	0	844	2505			603	1764		
CD45	758	7130	0	0	1103	3259			802	2395		
CD56	32	281	0	0	46	128			45	97		
CD68	551	1528	0	0	802	698			588	439		
CD69	744	3693	0	0	1083	1688			840	1091		
CD8	146	914	0	0	212	417			160	286		
area	0.69	2.19	0.00	0.00								

Table 57: SK2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1024	5210	0	0	1491	4700			1071	3221		
CD123	96	819	0	0	139	738			129	476		
CD15	344	834	0	0	500	752			395	370		
CD279	136	784	0	0	198	707			135	503		
CD3	343	4049	0	0	499	3653			422	2801		
CD4	580	4636	0	0	844	4183			603	3033		
CD45	758	5575	0	0	1103	5030			802	3651		
CD56	32	221	0	0	46	199			45	129		
CD68	551	1268	0	0	802	1144			588	700		
CD69	744	3230	0	0	1083	2914			840	1969		
CD8	146	825	0	0	212	744			160	547		
area	0.69	1.11	0.00	0.00								

Table 58: SK2_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	444	4568	0	0	740	1967			445	1368		
CD123	135	2885	0	0	225	1242			268	895		
CD279	34	124	0	0	56	53			67	34		
CD3	258	5477	0	0	430	2359			299	1779		
CD4	355	5541	0	0	592	2386			395	1759		
CD68	1047	1423	0	0	1746	613			1046	334		
CD68_raw	1946	3284	0	0	3245	1414			1775	799		
CD8	123	1305	0	0	205	562			139	400		
area	0.60	2.32	0.00	0.00								

Table 59: SK3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				Regularized Density			
	normal		lesion		normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	444	3423	0	0	740	2953			445	1855		
CD123	135	2425	0	0	225	2092			268	1540		
CD279	34	113	0	0	56	97			67	66		
CD3	258	4809	0	0	430	4149			299	3301		
CD4	355	4797	0	0	592	4139			395	3167		
CD68	1047	1314	0	0	1746	1133			1046	636		
CD68_raw	1946	2907	0	0	3245	2508			1775	1402		
CD8	123	1212	0	0	205	1045			139	801		
area	0.60	1.16	0.00	0.00								

Table 60: SK3_Les. All measurements are up to 500 um below DEJ. Area is measured in mm2. Density in count per mm2. Regularized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.